

CS & IT ENGINEERING



THEORY OF COMPUTATION

REGULAR EXPRESSION

Lecture No.- 01



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Recap of Previous Lecture



(2-3)

Topic

DFA Construction

?????

NFA

E-NFA



Topics to be Covered



Topic

Regular Expression Construction

Topic

?? Properties of Regular Expression

Topic

??

Finite Automata $\Leftrightarrow$ Regular Expression

Topic

??

Finite Automata $\Leftrightarrow$ Grammar $\Leftrightarrow$ Reg. Expr



Topic : Regular Expression

Language = $\{\epsilon, a, b, aa, ab, \dots\}$

$(a+b)^*$

$(a+b)^* ab (a+b)^*$

① The simplest way of representing a regular language is known as Regular expression.

$\checkmark L = \{a^n b^n \mid n \geq 0\}$ not possible

② For every regular language regular expression can be constructed.

③ To construct regular expression following 3 operators are used.

- $\{ + \}$ is known as union operator
- $\{ . \}$ is known as concatenation operator
- $\{ * \}$ is known as Kleene closure operator

$$\overset{(x)}{\gamma_1} + \overset{(y)}{\gamma_2} = \{\gamma_1, \gamma_2\}$$

$$\underline{\gamma_1} \cdot \underline{\gamma_2} = \{\gamma_1 \gamma_2\}$$

$$\gamma^* = \gamma^0 + \gamma^1 + \gamma^2 + \dots$$



Topic : NOTE



- ① For one regular language many number of regular expressions can be possible.
- ② One regular expression can generate only one regular language.

$$L_1 = \{ \} \longrightarrow \Phi \quad \text{Regular Expression}$$

$$L_2 = \{ \epsilon \} \longrightarrow \epsilon$$

$$L_3 = \{ a \} \longrightarrow a$$

$$L_4 = \{ a, b \} \longrightarrow a + b$$

$$L_5 = \{ aa, ab, ba \} \longrightarrow aa + ab + ba$$

$$L_6 = \{$$

$$L_6 = \{\epsilon, a^1, a^2, a^3, \dots\} \longrightarrow a^* \\ + \text{ [Positive closure]}$$

$$L_7 = \{a^1, a^2, a^3, a^4, \dots\} \longrightarrow a$$

$$L_8 = \{\epsilon, a, b, aa, ab, ba, bb, \dots\} \Rightarrow (a+b)^*$$

$$L_9 = \{a, b, aa, ab, ba, bb, \dots\} \Rightarrow (a+b)^+$$

$$L_{10} = \{a^n b^m \mid n \neq m\} \Rightarrow \underline{\underline{\text{not possible}}}$$

$$L_{11} = \{ \overset{a^*}{\underline{a^n}} \overset{b^*}{\underline{b^m}} \mid n, m \geq 0 \} \longrightarrow a^* b^*$$

$$L_{12} = \{ a^n b^m \mid n, m \geq 1 \} \longrightarrow a^+ b^+$$

$$L_{13} = \{ a^n b^m \mid \begin{matrix} n \geq 0 \\ m \geq 1 \end{matrix} \} \longrightarrow a^* b^+$$

$$L_{14} = \{ a^n b^m \mid \begin{matrix} n \geq 2 \\ m \geq 3 \end{matrix} \} \longrightarrow \underline{aaa^*} \underline{bbb^*}$$

$$L_{15} = \{ a^n (b^2)^m \mid n, m \geq 0 \} \longrightarrow \underline{a^* (bb)^*}$$

$a^* (bb)^*$

$$\begin{matrix} (a+b)^* \\ b \\ (ab)^* \\ \underline{abab} \end{matrix}$$

Possible

$$L_{16} = \{a^n b^m \mid (n+m) \text{ is even}\}$$

$$(aa)^*(bb)^* + a(aa)^*b(bb)^*$$

$$\begin{cases} \text{even} + \text{even} = \text{even} \\ \text{odd} + \text{odd} = \text{even} \end{cases}$$

$$L_{17} = \{a^n b^m \mid (n+m) \text{ is odd}\}$$

$$\left\{ \underbrace{(aa)^*}_{a^2} \underbrace{b(bb)^*}_{b^3} + \underline{a(aa)^*} \underline{(bb)^*} \right\}$$

$$L_{18} = \{a^n b^m \mid n > m\} \rightarrow$$

0000100

$$L_9 = \{1, 2, 4, 8, 16, \dots, 2^n, \dots\} = \boxed{0^*10^*}$$

All these numbers written in binary

$$L_{20} = \{1, 2, 4, 8, 16, \dots, 2^n, \dots\}$$

All these numbers written in Unary

not possible

Yes
No

$$(a+b)^* = (a+b)^0 + (a+b)^1 + (a+b)^2 + (a+b)^3 + \dots$$

$$= \epsilon + a + b + aa + ab + ba + bb + \dots$$

#Q. Construct regular expression that generates set of all strings of a's and b's where length of each string is exactly 4.

$$\frac{(a+b)}{a/b} \cdot \frac{(a+b)}{a/b} \cdot \frac{(a+b)}{a/b} \cdot \frac{(a+b)}{a/b}$$

#Q. Construct regular expression that generates set of all strings of a's and b's where length of each string is atleast 4

$$\left\{ \underline{(a+b)} \underline{(a+b)} \underline{(a+b)} \underline{(a+b)} (a+b)^* \right\}$$

$\{0, 1, 2, 3, 4\}$

#Q. Construct regular expression that generates set of all strings of a's and b's where length of each string is atmost 4.

$(a + b + \epsilon)(a + b + \epsilon)(a + b + \epsilon)(a + b + \epsilon)$

$\{\text{atmost } 4\}$

#Q. Construct regular expression that generates set of all strings of a's and b's where length of each string is divisible by 4.

$$\underline{[(a+b)^4]^*}$$

$$\Sigma = \{a, b\}$$

#Q. Construct regular expression that generates set of all strings of a's and b's where number of a's are exactly 4

$$b^* \underline{a} b^* \underline{a} b^* \underline{a} b^* \underline{a} b^*$$

#Q. Construct regular expression that generates set of all strings of a's and b's where number of a's are atleast 3.

#Q. Construct regular expression that generates set of all strings of a's and b's where number of a's are atmost 3.

$$b^* \underline{(a+e)} b^* \underline{(a+e)} b^* \underline{(a+e)} b^*$$

bbbbb

#Q. Construct regular expression that generates set of all strings of a's and b's where number of a's are divisible by 3.

$$\left[b^* a b^* a b^* a b^* \right]^* + b^*$$



#Q. How many states are there in minimal DFA that accept following regular expression.

#Q. Construct regular expression that generates set of all strings of a's and b's where each string starting and ending with different symbol.

#Q. Construct regular expression that generates set of all strings of a's and b's where having substring aab.

#Q. Construct regular expression that generates set of all even length palindrome strings over {a}.

$$\{\epsilon, a^2, a^4, a^6, \dots\} = (aa)^*$$

→ madam ←

→ malayalam ←

#Q. Construct regular expression that generates set of all even length palindrome strings over $\{a\}$.

#Q. Construct regular expression that generates set of all even length palindrome strings over $\{a, b\}$.

$\{\epsilon, aa, bb, abba, baab, \dots\}$

not possible

#Q. Construct regular expression that generates set of all odd length palindrome strings over $\{a, b\}$.

#Q. Construct regular expression that generates set of all odd length palindrome strings of English language.

not possible



Topic : NOTE



$\frac{W}{a} \quad \frac{W^R}{a}$

- Palindrome languages over more than one symbol are not regular .Hence regular expression not possible.
- Palindrome languages over one symbol are regular.

$\frac{W}{mad} \quad \frac{W^R}{am}$

#Q. Construct Regular Expression for the following languages

① Peter Linz

② Ullman

THANK - YOU